

Metabolic dysfunction-associated steatotic liver disease - A new indication for sodium-glucose Co-transporter-2 inhibitors.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) has been proposed as a new name for the previous non-alcoholic fatty liver disease (NAFLD). There are some differences between MASLD and NAFLD, e.g., diagnostic criteria. MASLD is a hepatic steatosis without harmful alcohol consumption and is caused by metabolic factors. The prevalence of MASLD varies amongst different populations. The change in lifestyle plays a fundamental role in MASLD management, while there is no registered pharmacotherapy in this indication. Sodium-glucose co-transporter 2 inhibitors (SGLT2i) have been suggested to have a beneficial effect on hepatic steatosis, hence, they have been widely investigated as potential therapeutics in MASLD. In this review, we aimed to thoroughly summarize current evidence from original research about the effects of SGLT2i use on MASLD. Almost all discussed studies advocate using SGLT2i in MASLD because of their beneficial effects. It includes the loss of body weight, which is beneficial per se, and the improvement in hepatic parameters. Most importantly, steatosis reduction has been observed in patients using SGLT2i. We highly recommend further research in this field, which we believe will eventually lead to a new indication for SGLT2i, i.e., MASLD.

Relationship between sarcopenia and metabolic dysfunction-associated steatotic liver disease (MASLD): A systematic review and meta-analysis.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) formerly known as Nonalcoholic fatty liver disease (NAFLD) is a common chronic disease. Identifying MASLD risk factors could help early intervention and reduce the burden of the disease. Previous studies investigated the association between sarcopenia and NAFLD. Several trials were published after the last meta-analysis with indecisive results. This is an updated meta-analysis which aims to assess the association between sarcopenia, MASLD, and MASLD-related fibrosis. Relevant trials published on PubMed, Web of Science, Scopus, and Cochrane Library databases until October 2022 were included. We included studies in which skeletal mass index (SMI) or sarcopenia was compared between patients with and without NAFLD now MASLD. Also, studies comparing fibrosis between MASLD patients with and without sarcopenia were included. Data were pooled as odds ratios (ORs) and 95 % confidence intervals (CIs) using Review Manager Software. A total of 25 studies were included. The incidence of sarcopenia was significantly higher in MASLD than controls (OR, 1.25; 95 % CI, 1.08-1.44; $P = 0.003$). SMI odds showed no significant difference between MASLD patients and controls (OR, 1.02; 95 % CI, 0.91-1.15; $P = 0.7$). MASLD patients with sarcopenia had higher odds of fibrosis than MASLD patients without sarcopenia (OR, 1.49; 95 % CI, 1.03-2.14; $P = 0.03$). Sarcopenia increased MASLD's probability and was associated with a higher probability of liver fibrosis in MASLD patients. However, SMI had no predictive value of MASLD occurrence.

EASL-EASD-EASO Clinical Practice Guidelines on the Management of Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed non-alcoholic fatty liver disease (NAFLD), is defined as steatotic liver disease (SLD) in the presence of one or more cardiometabolic risk factor(s) and the absence of harmful alcohol intake. The spectrum of MASLD includes steatosis, metabolic dysfunction-associated steatohepatitis (MASH, previously NASH), fibrosis, cirrhosis and MASH-related hepatocellular carcinoma (HCC). This joint EASL-EASD-EASO guideline provides an update on definitions, prevention, screening, diagnosis and treatment for MASLD. Case-finding strategies for MASLD with liver fibrosis, using non-invasive tests, should be applied in individuals with cardiometabolic risk factors, abnormal liver enzymes, and/or radiological signs of hepatic steatosis, particularly in the presence of type 2 diabetes (T2D) or obesity with additional metabolic risk factor(s). A stepwise approach using blood-based scores (such as FIB-4) and, sequentially, imaging techniques (such as transient elastography) is suitable to rule-out/in advanced fibrosis, which is predictive of liver-related outcomes. In adults with MASLD, lifestyle modification - including weight loss, dietary changes, physical exercise and discouraging alcohol consumption - as well as optimal management of comorbidities - including use of incretin-based therapies (e.g. semaglutide, tirzepatide) for T2D or obesity, if indicated - is advised. Bariatric surgery is also an option in individuals with MASLD and obesity. If locally approved and dependent on the label, adults with non-cirrhotic MASH and significant liver fibrosis (stage ≥ 2) should be considered for a MASH-targeted treatment with resmetirom, which demonstrated histological effectiveness on steatohepatitis and fibrosis with an acceptable safety and tolerability profile. No MASH-targeted pharmacotherapy can currently be recommended for the cirrhotic stage. Management of MASH-related cirrhosis includes adaptations of metabolic drugs, nutritional counselling, surveillance for portal hypertension and HCC, as well as liver transplantation in decompensated cirrhosis.

Management of Metabolic-Associated Fatty Liver Disease/Metabolic Dysfunction-Associated Steatotic Liver Disease: From Medication Therapy to Nutritional Interventions.

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Non-alcoholic fatty liver disease (NAFLD) is a common long-lasting liver disease that affects millions of people around the world. It is best identified with a hepatic fat build-up that ultimately leads to inflammation and damage. The classification and nomenclature of NAFLD have long been a controversial topic, until 2020 when a group of international experts recommended substituting NAFLD with MAFLD (metabolic dysfunction-associated FLD). MAFLD was then terminologically complemented in 2023 by altering it to MASLD, i.e., metabolic dysfunction-associated steatotic liver disease (MASLD). Both the MAFLD and the MASLD terminologies comprise the metabolic element of the disorder, as they offer diagnostic benchmarks that are embedded in the metabolic risk factors that underlie the disease. MASLD (as a multisystemic disease) provides a comprehensive definition that includes a larger population of patients who are at risk of liver morbidity and mortality, as well as adverse cardiovascular and diabetes outcomes. MASLD highlights metabolic risks in lean or normal weight individuals, a factor that has not been accentuated or discussed in previous guidelines. Novel antihyperglycemic agents, anti-hyperlipidemic drugs, lifestyle modifications, nutritional interventions, and exercise therapies have not been extensively studied in MAFLD and MASLD. Nutrition plays a vital role in managing both conditions, where centralizing on a diet rich in whole vegetables, fruits, foods, healthy fats, lean proteins, and specific nutrients (e.g., omega-3 fatty acids and fibers) can improve insulin resistance and reduce inflammation. Thus, it is essential to understand the role of nutrition in managing these conditions and to work with patients to develop an individualized plan for optimal health. This review discusses prevention

strategies for NAFLD/MAFLD/MASLD management, with particular attention to nutrition and lifestyle correction.

Steatotic liver disease.

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Steatotic liver disease is the overarching term for conditions characterised by abnormal lipid accumulation in the liver (liver or hepatic steatosis). Steatotic liver disease encompasses what was previously termed non-alcoholic fatty liver disease (NAFLD), which is now called metabolic dysfunction-associated steatotic liver disease (MASLD). Additionally, steatotic liver disease includes alcohol-related liver disease (ALD) and MetALD, the new classification for the overlap between MASLD and ALD, and rare causes of liver steatosis. Cirrhosis is globally the 11th leading cause of death, and steatotic liver disease has become the leading cause of cirrhosis in the EU and USA. Steatotic liver disease affects around 30% of the global population and is mainly driven by obesity, type 2 diabetes, and alcohol intake, but only a minor proportion with steatotic liver disease progress to cirrhosis. The presence and progression of liver fibrosis led by hepatic inflammation is the main predictor of liver-related death across the entire spectrum of steatotic liver diseases. A combination of recent advancements of widely available biomarkers for early detection of liver fibrosis together with considerable advancements in therapeutic interventions offer the possibility to reduce morbidity and mortality in patients with steatotic liver disease. This Seminar covers the recent reclassification of steatotic liver disease and how it reflects clinical practice and prognosis. For early detection of liver fibrosis, we propose a collaborative diagnostic framework between primary care and liver specialists. Lastly, we discuss current best practices for managing steatotic liver disease, we explore therapeutic targets across the spectrum of steatotic liver diseases, and we review the pipeline of drugs in development for MASLD.

Why MASLD Lags Behind MAFLD: A Critical Analysis of Diagnostic Criteria Evolution in Metabolic Dysfunction-Associated Liver Diseases.

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Emerging in the 1800s under the label "fat in the liver" and later gaining prominence in the 1980 as non-alcoholic fatty liver disease (NAFLD), the disease predominantly attributed to metabolic dysfunction presents a formidable health issue marked by substantial morbidity and mortality. It was 2020 when a change of one letter "NAFLD" to metabolic dysfunction-associated fatty liver disease "MAFLD" linked with the change in the definition and diagnostic criteria began a new controversy around the globe. Metabolic dysfunction-associated fatty liver disease (MAFLD) criteria represent a substantial departure from previous diagnostic measures of NAFLD, and provide the first set of positive criteria for diagnosis of the disease in adults and children that emphasise the key attribute of metabolic dysfunction in the pathogenesis, and acknowledges that the disease is a continuum across the life span. In 2023, an adapted version of the diagnostic criteria of MAFLD was proposed to define a slightly modified term; metabolic dysfunction-associated steatotic liver disease (MASLD). The MASLD criteria did not provide any conceptual advantage, and emerging evidence suggests that it actually performs worse than the MAFLD criteria. This raises the intriguing question of why MASLD was unable to take advantage of being second? In this review, we will explore the possible reasons for this unique case and highlight the current

evidence supporting the use of MAFLD instead of MASLD in defining metabolic dysfunction-associated fatty liver diseases.

Phytonanotherapy for the Treatment of Metabolic Dysfunction-Associated Steatotic Liver Disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as nonalcoholic fatty liver disease, is a steatotic liver disease associated with metabolic syndrome (MetS), especially obesity, hypertension, diabetes, hyperlipidemia, and hypertriglyceridemia. MASLD in 43-44% of patients can progress to metabolic dysfunction-associated steatohepatitis (MASH), and 7-30% of these cases will progress to liver scarring (cirrhosis). To date, the mechanism of MASLD and its progression is not completely understood and there were no therapeutic strategies specifically tailored for MASLD/MASH until March 2024. The conventional antiobesity and antidiabetic pharmacological approaches used to reduce the progression of MASLD demonstrated favorable peripheral outcomes but insignificant effects on liver histology. Alternatively, phyto-synthesized metal-based nanoparticles (MNPs) are now being explored in the treatment of various liver diseases due to their unique bioactivities and reduced bystander effects. Although phytonanotherapy has not been explored in the clinical treatment of MASLD/MASH, MNPs such as gold NPs (AuNPs) and silver NPs (AgNPs) have been reported to improve metabolic processes by reducing blood glucose levels, body fat, and inflammation. Therefore, these actions suggest that MNPs can potentially be used in the treatment of MASLD/MASH and related metabolic diseases. Further studies are warranted to investigate the feasibility and efficacy of phytonanomedicine before clinical application.

Impact of PNPLA3 in Lean Individuals and in Cryptogenic Steatotic Liver Disease.

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Although metabolic dysfunction-associated steatotic liver disease (MASLD) is strongly associated with obesity, around 20% of individuals with hepatic steatosis may nonetheless have normal body mass index, a condition often referred to as lean nonalcoholic fatty liver disease (NAFLD). Under the new nomenclature and definition of MASLD, lean NAFLD can be further divided into lean MASLD (when there is one or more cardiometabolic risk factors) and cryptogenic steatotic liver disease (when there is no cardiometabolic risk factor). Current studies suggest that the at-risk PNPLA3 rs738409 variant is more common among individuals with lean NAFLD than their overweight and obese counterparts. However, even in this group, cardiometabolic risk factors are often required for the development of hepatic steatosis and liver injury. In the general population, PNPLA3 gene polymorphism is associated with an increased risk of MASLD, more severe liver histology (i.e., the presence of steatohepatitis and fibrosis) and future development of hepatocellular carcinoma and cirrhotic complications. Emerging data also suggest that individuals carrying the PNPLA3 GG genotype might have a greater reduction in hepatic steatosis and liver enzymes with lifestyle intervention and metabolic treatments, such as glucagon-like peptide-1 receptor agonists. Studies have not specifically examined the impact of PNPLA3 in lean individuals.

Clinical outcomes of MAFLD versus NAFLD: A meta-analysis of observational studies.

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The recent change in terminology from nonalcoholic fatty liver disease (NAFLD) to metabolic dysfunction-associated fatty liver disease (MAFLD) and metabolic dysfunction-associated steatotic liver disease (MASLD) highlights the link between hepatic steatosis and metabolic dysfunction, taking out the stigmata of alcohol. We compared the effects of NAFLD and MAFLD definitions on the risk of overall and cardiovascular (CV) mortality, liver-related events (LRE), nonfatal CV events (CVE), chronic kidney disease (CKD), and extra-hepatic cancers (EHC). We systematically searched four large electronic databases for cohort studies (published through August 2023) that simultaneously used NAFLD and MAFLD definitions for examining the risk of mortality and adverse CV, renal, or oncological outcomes associated with both definitions. In total, 21 eligible cohort studies were identified. Meta-analysis was performed using random-effects modelling. Compared with those with NAFLD, individuals with MAFLD had significantly higher rates of overall mortality (random-effect OR 1.12, 95% CI 1.04-1.21, $p = .004$) and CV mortality (random-effect OR 1.15, 95% CI 1.04-1.26, $p = .004$), and a marginal trend towards higher rates of developing CKD (random-effect OR 1.06, 95% CI 1.00-1.12, $p = .058$) and EHC events (random-effect OR 1.11, 95% CI 1.00-1.23, $p = .052$). We found no significant differences in the risk LREs and nonfatal CVE between MAFLD and NAFLD. Meta-regression analyses identified male sex and metabolic comorbidities as the strongest risk factors related to the risk of adverse clinical outcomes in MAFLD compared to NAFLD. Individuals with MAFLD have higher rates of overall and CV mortality and higher rates of developing CKD and EHC events than those with NAFLD, possibly due to the dysmetabolic risk profile related to MAFLD.

Comparing similarities and differences between NAFLD, MAFLD, and MASLD in the general U.S. population.

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Recently, the multisociety Delphi consensus renamed non-alcoholic fatty liver disease (NAFLD) terminology [previously renamed metabolic-associated fatty liver disease (MAFLD)] as metabolic dysfunction-associated steatotic liver disease (MASLD). The aim of this study was to compare the similarities and differences between NAFLD, MAFLD, and MASLD and to clarify the impact of this new name change. A cross-sectional study of 3,035 general subjects with valid vibration-controlled transient elastography data was conducted based on data from the National Health and Nutrition Examination Survey (NHANES) 2017-2020. NAFLD, MAFLD, and MASLD were defined according to the corresponding consensus criteria. Using controlled attenuation parameter (CAP) ≥ 274 dB/m and liver stiffness measurements (LSM) ≥ 9.7 kPa as the cutoff values for the presence of hepatic steatosis and advanced liver fibrosis (ALF), the prevalence of NAFLD, MAFLD, and MASLD were 38.01% (95% CI 35.78-40.29%), 41.09% (39.09-43.12%), and 37.9% (35.70-40.14%), respectively, and the corresponding prevalence of ALF was 10.21% (7.09-14.48%), 10.13% (7.06-14.35%), and 10.24% (7.11-14.53%), respectively. The kappa values for the three definitions were above 0.9. The prevalence and severity of the three definitions remained similar when the sensitivity analyses were performed using different CAP thresholds. The prevalence of NAFLD, MAFLD, MASLD, and ALF increased as the number of cardiometabolic risk factors (CMRF) increased. Our findings highlight the consistency among the three definitions, especially between NAFLD and MASLD, so that the new consensus will not disturb the original NAFLD-related findings. Additionally, more attention should be paid to patients with a high number of CMRFs.
